

# Exploring Esports Spectator Motivations

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## ABSTRACT

Gameplay spectatorship has developed significantly over the last decade, necessitating the need to design more engaging spectator experiences. In order to do this effectively, however, we must first better understand spectator behavior and motivations. In this paper, we build on the results of a previous work examining interaction preferences of remote livestream spectators depending on their user type, and report a new qualitative study exploring the underlying motivations that shape the experience of those engaged in interactive spectating. Our results highlight five main themes (entertainment, team support, learning, caster, social) that motivate spectators to engage with interactive esports experiences. This work will motivate ongoing research in this domain by promoting conversations about livestream spectator engagement and motivations, and facilitating engaging spectator experiences.

## CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Interaction design**.

## KEYWORDS

esports, livestreaming, thematic analysis, motivations, spectator typology, Twitch

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## 1 INTRODUCTION

Evaluating and understanding the player experience (PX) is a critical undertaking in game development. While current research often

examines users directly interacting with the game content (players), there is a notable cohort that do not only consume video games in the traditional way: video game stream spectators. Our research aims to study user behaviour and design considerations specifically for interactive spectatorship (audience members that engage with content at a much deeper level, participating actively [17]).

Gameplay spectatorship has developed significantly over the last decade, with esports spectatorship expected to grow to 577 million viewers by 2024 [13]. Much of this spectatorship occurs on platforms designed for streaming gameplay, such as Twitch.tv (currently the most popular streaming platform)—as well as more recent competitors, such as Facebook Live and YouTube Streaming [13]. In this domain, one design opportunity is to improve spectator engagement by adding new features, or enhancing existing features such as those described by Stahlke et al [17]) to enable interaction with streamers, players, or other spectators. Our research focuses on designing and evaluating interactive spectator experiences. This Late-Breaking Work (LBW) describes the current stage of our project on developing a spectator-specific user typology that aims to help developers better understand behaviour and its underlying motivation of their target audience. By contributing an understanding of spectator motivations and behaviours, we can equip stakeholders (e.g., developers, stream service providers) with the tools and knowledge required to broadly improve spectator experiences (thus encouraging engagement and retention, while simultaneously offering experiential benefits for both spectators and streaming communities).

In this paper, we build on the results of a previous survey discussing the interaction preferences of livestream spectators depending on their user type [21], and report a new qualitative study exploring the underlying motivations that shape the experience of those engaged in interactive spectating. Our results highlight five main themes (entertainment, team support, learning, caster, social) that motivate remote livestream spectators to engage with interactive esports experiences. This work will motivate ongoing research in this domain by promoting conversations about spectator engagement, and facilitating engaging spectator experiences.

## 2 BACKGROUND

In recent years, esports has become a widespread phenomenon [7]. Just as in live sporting events, spectating is an integral part of the esports experience. Spectating video gaming, however, while having a small physical audience at the live event, is largely accessible

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to audiences through online streaming platforms like Twitch or YouTube Live, which enable a widespread remote audience to participate. Twitch in particular has become the leading platform for gameplay streaming, popularized not only for esports broadcasts, but also for hobbyist video game streamers. Interactive features that are unique to these livestreaming platforms (e.g. cheering, betting or voting) have evolved in recent years, allowing for more dynamic and rich spectator experiences [14, 16]. These platforms also allow spectators to interact with one another through a live-chat feature that exists alongside the game, allowing spectators to chat with one another, and the streamer, in real time.

Attempts have been made within the field of game user research (GUR) to understand player/streamer motivations [22], but less work has been done to understand spectators, to better design for audience experiences [8]. Striner et al. attempt to classify the interactive audience experience, creating a spectrum of audience interactivity from “observing passively” to “becoming the performer” [18]. In 2020, Schofield et al. discussed motivations of content creators and viewers on the Twitch platform [15], and another study found that social motivations are a large component motivating stream watchers [9].

However, the realm of esports differs from general livestreams in that it potentially consists of millions of spectators [6, 11], which changes the way spectators interact with others or their behaviour in the live chat (e.g. spamming memes, cheering as a crowd rather than having meaningful conversation with other spectators due to the sheer amount and speed of which inputs are being displayed at any given moment.) [4, 5]. Some work has been undertaken in an effort to understand game spectator motivations: one study found that while on-site attendees are motivated by social factors, this is not the case with live stream spectators, as they are particularly interested in gaining knowledge and skills [12]; this was supported by other work that showed “informational motivation were found to be the strongest positive predictor of the use of streaming platforms” [3]. Despite each of these attempts to categorize spectator motivations, there is still not yet a consensus on the motivations of streamers, and how they relate to a particular typology system.

In contrast to understanding spectator motivations, understanding *player* motivations has long been a topic of interest in GUR. In 1996, Bartle et. al created an early taxonomy of player motivations, dividing them based on their interests and motivations [1]. Later, Marczewski developed the Gamification User Type Hexad framework [10], which can be described as a generalized taxonomy based on Bartle’s initial model, and Yee et. al discuss the three main components of the original taxonomy in their work [19, 20]. However, when turning to spectator motivations, the research is comparatively limited. Cheung and Huang represent one of the only entries to the literature exploring this topic: by examining facial expressions of fans during tournaments, they categorize spectators according to nine personas to help understand why spectators watch, and how the various stakeholders affect the overall experience. Our work expands on this work and the work of Yu et. al [21], which profiles livestream spectators, uncovering possible trends in spectators’ interaction preferences depending on their user type (as it relates to the player Hexad scale [10]).

### 3 STUDY DESIGN

Our key research question is to study general user motivations in spectating esports (Q1), as well as what motivates spectators to use interactive features in online livestreaming platforms (Q2). Answering these research questions would support our goal to understand the viewing preferences and motivations of esports tournament spectators.

#### 3.1 Methods

We administered an online survey through Google Forms in June of 2021. League of Legends was chosen as the title of the study due to its widespread popularity as well as its active and large esports fan base. Upon providing consent, participants filled out the survey consisting of general demographic information, general questions to justify inclusion (e.g. “how often do you watch League of Legends?”), and four open-ended questions asking about viewer preferences and habits. These questions ask the participants to recall their experiences while watching esports. The first item asked the participants about their general experience while watching esports tournaments, while subsequent questions asked about their experiences, feelings, and other activities they were involved in while spectating tournaments. The questions were formed through numerous expert discussions with all contributing authors.

#### 3.2 Participants

A total of 56 participants responded to the survey. In order to participate, participants were required to have watched a professional League of Legends tournament at least once per month during tournament season. Participants were recruited using Twitter and the subreddits r/LeagueOfLegends and r/lolesports. These social media platforms were chosen because of their large and active user base, and their accessibility to our target audience (i.e., LoL spectators). Participant ages ranged from 19 to 30 years of age ( $M=23.57$ ;  $SD=4.44$ ). 85% of participants responded that they play League of Legends at least once a week while 43% of the participants play almost every day. Most of the participants also are frequent tournament watchers. 90% of the participants responded that they watch tournaments at least once a month, while 23% of the participants would watch whenever there is a game.

#### 3.3 Analysis

All responses received from the survey were analyzed qualitatively using thematic analysis, following Braun and Clarke’s guidelines for inductive thematic analysis [2]. Each statement was tagged as one theme based on the sentiments of the statement. The resulting codebase was then processed based on their frequency of appearance and association with participants’ other answers.

To analyze the data that we have collected, an open source qualitative research software called Taguette<sup>1</sup> was used to assist processing of the data. All the responses were exported from Google forms to a datasheet. The data was then imported into Taguette. Each statement provided by the participants was then given a tag word based on the sentiment of the statement. For example: “I enjoy games where my favourite team is playing. I support them win or

<sup>1</sup><https://www.taguette.org/> Accessed: June 2021

lose." would be given the tag "support". Similarly statements such as: "I don't like interacting with others in chat. Reading them is fun sometimes, but I don't participate myself." would be given the tag "passive".

Tags are determined through a filtering process. Initially a set of tags were determined based on expectations of what we think might appear. During the tagging process, if no predetermined tags were able to represent the sentiment of the statement, a new tag would be added. After all statements were tagged, all tags were then reviewed and any tags that represent the same sentiment would then be merged to the tag that best describes the sentiments. This process was done using Google sheets. The final step was to count the appearance of each tag.

## 4 RESULTS

In this section, we highlight key results gathered from the exploratory survey. To summarize the findings, when asked about the central elements of their general spectating motivations, the most popular sentiments were strategy/gameplay (28%), support (23%), competitive environment (20%), and excitement/hype (14%). Almost half of the participants (25) stated that they are not motivated to use interactive features for reasons such as: "I don't use chat" (P45), "Well i never used those so i can't really tell what would motivate me. Maybe if the discussion felt like it was in a smaller circle of people rather than 50 000" (P49). Among those that do, the most popular sentiments are social aspects (25%), entertainment (16%), and incentives/reward (9%).

### 4.1 General Experience

'General Experience' explores the general motivations reported by participants for engaging in online spectatorship. We prompted participants to reflect on their overall experience while watching professional esports tournaments. In general, participants provided a variety of reasons for their engagement in esports spectatorship, which have been categorized into five recurring themes. These factors are not mutually exclusive and may co-occur to influence the spectating experience as participants often experience a preference for multiple themes. The following section details the themes and provides supporting participant quotes, which are reported verbatim.

**Entertainment:** Over half of the participants indicated that their primary motivation for tuning into professional esports livestreams is to have an entertaining experience. For instance, P14 states that, "I watch for entertainment...I feel excitement from new strategies or hype plays," while P22 claims, "I usually experience excitement when games are close." Participants also alluded to the sentiment that they watch esports for the same entertainment value provided by traditional sports: "It's the same as watching sports. The gameplay is really impressive and there is a lot of emotional highs every game" (P37), and "I watch it [esports] the same I watch a football match or a movie" (P3).

**Support:** Participants expressed a motivation to watch and support their favourite organisation, players, or geographic region. Participants stated that, "I enjoy cheering for my favourite team" (P4), "I also really like specific players, and always hope they do well in the game" (P25), and that they spectate due to an "investment

in team" (P6). Interestingly, motivation to spectate was dampened by a lack of success from a supported team or region: "I haven't watched a tournament in years because I live in NA and our region has done poorly at the world tournament... the repeated bad results took away the anticipation" (P8).

**Learning:** Participants highlighted learning gameplay strategies from observing professional play as a motivating factor in their engagement. For instance, P12 states that, "It's also fun to analyze macro and see how/why teams use their players, map advantage and tempo to get advantages or waste it." Additionally, participants tune into livestreams to "learn about technique" (P6), and to "learn new strategies" (P15).

**Caster<sup>2</sup>/Analysts<sup>3</sup>:** Participants indicated engaging in spectatorship to consume content from a particular caster or analyst. For instance, P12 states that they enjoy "analysts breaking down the game and shoutcasters casting the teamfights." In particular, participants express that the livestream is more engaging when "the castors and analyst are good" (P1). P5 echoes this sentiment in the following statement: "The main issue I have with esports specifically with NA is the analyst desk. When I watch LEC for example, the personalities on it just seem more invested in their job and so this makes it much more fun to listen to them in comparison to the LCS."

**Social:** Socializing with others on the livestreaming platform (i.e., chat) as well as with friends was identified as a motivating factor for spectatorship. P20 alludes to spectatorship as a 'communal experience' in which gameplay is meant to be enjoyed with others—"...watching a great game by oneself is not nearly as interesting as watching a mediocre team with a community." Additionally, P7 states that "It [spectating] ends up being a social thing, watching to be able to bring it up with friends later." Participants also reported "interacting with community in chat" (P9), typically by spam—i.e., "spamming the chat with cypypasta" (P15), "spamming BASED in chat" (P19), and spamming "PogPogPogu in da chat" (P9)—as "Twitch chat is a lot of fun and copy-pastas are the biggest part of big tournaments" (P22).

When looking at detailed answers, some of the most popular sentiments are often paired together. For example: "I really like weird games, where teams try to innovate with the champions they pick or strategies they choose for the game (like different builds, lane swaps, jungle pathing,...). I also really like specific players, and always hope they do well in the game. For example, if one of the players gets a solo kill in lane, or was a big reason why his team won a teamfight, it will make me happy." (P25). This and similar longer statements would typically all express a combination of innovative strategies, support for specific players, the competitions, as well as how it affects them emotionally.

Shorter statements are more explicit with their sentiments such as the most memorable experiences are when their supporting parties are at play. For example: "Watching the individual players." (P47), "when your (my) favorite team plays an important match" (P48). The excitement when the game is competitive, "Crucial moments in the game when anything can happen. Turnaround events and such that get the blood pumping." (P49). Unique answers that reflect social engagement such as "discussing in-game choices and impacts with community" (P20). There are also expected popular answers that

<sup>2</sup>Casters are commentators that describe live as well as comments on the live match.

<sup>3</sup>Analysts are commentators that make predictions and summaries of the match usually before and after the match.

| Motivation              | Description                                                                   |
|-------------------------|-------------------------------------------------------------------------------|
| Entertainment           | Spectates for entertainment, enjoyment, and fun                               |
| Support                 | Spectates to support a player, team or region                                 |
| Learning                | Spectates to learn skills and strategies, and translate learnings to gameplay |
| Caster/Analyst          | Spectates to watch/support a specific caster or analyst                       |
| Social                  | Spectates socially with others; social event                                  |
| Strategy/Gameplay       | Spectates to admire professional play                                         |
| Competitive environment | Spectates to enjoy the competitive and unpredictable nature of esports        |
| Excitement/Hype         | Spectates to join the excitement of the competition and the crowd             |
| Incentives/Rewards      | Spectates for extrinsic reward, such as in-game objects or currency           |
| Boredom                 | Spectates purely because nothing else interesting is happening                |

**Table 1: Spectators' motivations for watching League of Legends**

turned out to be quite rare such as, *"I like the twitch chat spamming memes"* (P1) and *"Analysts breaking down the game and shoutcasters casting the teamfights"* (P12). These comments are reflecting on what chat behaviours and caster performance respectively.

The overall sentiment of spectators' general experience reflects preference with a mix of key elements of a competitive environment where both teams display good strategy and gameplay. Another key element that seems to be "support". A portion of the participants had mentioned supporting their favourite teams or players when asked what it was that made it memorable.

## 4.2 Motivations to Use Interactive Features

'Motivations to Use Interactive Features' explores the motivations provided by participants concerning their engagement with online interactive features (e.g., chat, donations, polls) in livestreaming platforms—or their lack thereof. We asked participants to reflect on how they engage with the livestreaming platform and to elaborate on their experience. In this analysis, we identified three primary themes and two less prominent, yet noteworthy, themes. These themes are detailed, and supported by participant quotes in the following section:

*Social aspects:* Participants alluded to social aspects when referring to their motivations for using interactive features. For instance, some participants reported a social obligation to interact with such features—i.e., *"Friends using them"* (P29), and *"I may have sent a congrats to the particular streamer on their stream after the tournament ended"* (P8). Participants also reported engaging in interactive features outside of the livestreaming platform, such as Reddit and Discord. For instance, P7 states that, *"I read Reddit for the dumb comments and copy pastas,"* while P25 claims *"... the only time I'll voice my opinion about the game is if a player I like does really well, or one of the teams use a new or unpopular strategy. The ONLY way I'll ever do this is by commenting on Reddit."* Furthermore, P30 reports socially engaging in external platforms by *"Watching Worlds 2020 with a friend via screen sharing on Discord."*

*Entertainment:* Interactive features were reported to be a source of entertainment for participants. While many participants did not use chat or discussion features themselves, they enjoyed observing the interactions among chat users: *"I like looking at other people's reactions to certain plays or moments during the game even though I do not engage with it myself"* (P32), and *"...You almost get a feeling that*

*you are in stadium with all the people in chat"* (P22). Alternatively, participants reported *"clicking polls"* (P23) and *"betting with channel points"* (P56) as a source of entertainment within the livestream.

*Do not use features:* The majority of the participants (77%) reported that they typically did not use any interactive features. Participants refrain from engaging with interactive features as they are primarily *"focusing on watching the match"* (P46) or they *"don't see the point interacting with tournament livestreams because you get no input back except from other viewers"* (P8). Additionally, P38 claims their lack of participation in such features is attributed to the large viewer base: *"...the chat is simply to fast, and the only productive thing you can type are emotes reacting to whatever happens in the stream, which i don't usually do."* In line with this sentiment, P53 stated a preference for *"having a smaller community"* to interact with as it *"felt more involved."*

*Incentives:* Although less prominent, prize incentives were referenced as a motivator for using interactive features. For instance, P36 claimed that they engaged in interactive feature as *"there were prizes involved,"* while P8 stated that, *"...if the prize is good enough I will put a little effort into getting it, but I ignore small prizes because the games are not enjoyable enough to me."* As such, the reward must match the perceived effort from the spectator to promote engagement. Additionally, participants seek out external opportunities to gain rewards: *"The 'World's Pickem' Riot put out on esports.com to guess the tournament bracket for worlds is pretty popular amongst me and my friends on discord"* (P8).

*Hate watch:* Interestingly, some participants reported engaging in livestreams simply to observe others fail (i.e., 'hate watching'). For instance, P9 claimed to enjoy *"Watching pros play as an armchair andy laughing at how bad they are whilst simultaneously being way worse at the game is bare fun,"* while P46 enjoyed *"[watching] my favorite team (C9) smash Afreeca at worlds to advance to semis."* Although losing is an inevitable outcome for a team in a competitive match, fans often derive pleasure from observing the loss of an opposing team so that theirs may prevail.

## 5 DISCUSSION AND FUTURE WORK

This study generates insights into the motivations of livestream spectatorship—more specifically, livestreamed esports tournament spectatorship—in service of motivating the creation of an esports spectator typology. Our results indicate that spectators' motivations

differ substantially from those of traditional player motivations—establishing that co-opting a player typology system is not ideal, potentially in part due to interactions and interactive features that are unique to livestreaming environments (e.g. live chat stream, passive consumption of tournament livestreams). While a livestream spectator typology can not be derived directly from the results of this study, our findings help to inform future directions in establishing a typological framework. In this section, we discuss the implications of our data, suggest possible future routes for encouraging spectator engagement, and speculate potential new user profiles.

Notably, one popular motivation identified was the competitive environment. It is likely that spectators find reassurance in finding the team they associate with is better than other teams. In line with this sentiment, spectators may engage in ‘hate watching’ (i.e., a form of viewership where entertainment is derived from imposing criticism on disliked content or others) to reaffirm their beliefs about a team they deem inferior to that in which they support. Similarly, spectators may also experience pleasure (i.e., *schadenfreude*) from observing the failures of an opposing team.

Another prominent motivation to watch and interact with livestream tournaments is the desire to “support” one’s favourite team or players. One possible avenue of exploration is whether or not the motivation for socialization, or relatedness, may be met through other representation—that is, supporting a preferred player, team, or region. However, it may also be that watching tournaments because of a favourite team is simply a commonality amongst dedicated spectators.

One avenue for future research may be investigating how fan loyalty dictates stream viewing behaviour: for example, whether spectators watch solely because a preferred team or player is playing; or, if spectators are more likely to engage with interactive features when a preferred team is playing. Further, the nature of the match itself may motivate spectators to watch—it is possible that matches between rival teams, or games forecast to be closely matched, may influence spectator behaviour and motivations.

Further, “Entertainment” was identified as a motivating factor among returning spectators. The unpredictable nature of esports renders esports viewership a fun and entertaining spectating experience. To build off this experience, Riot provides spectators an opportunity to earn rewards (e.g., Blue Essence, Hextech chests, ultimate skins) through ‘Pickems’ by accurately predicting the winning teams in the World Championship Series. However, extrinsic rewards are not a central component of this feature as they are difficult to acquire; rather, spectators engage with this feature as it is entertaining to simply observe the outcome of a prediction. Moreover, some participants expressed incentives/rewards as a motivating factor in their engagement—an aspect that may be associated with entertainment value. As such, a similar approach could be implemented in online streaming platforms (for those whose entertainment is derived from extrinsic rewards) by utilizing channel points and Prediction polls, whereby players earn rewards by accurately predicting a gameplay event, such as “which team will get first blood.”

Turning to solutions for encouraging engagement in livestreams: in order to further facilitate support and social interactions among

the livestreaming communities, we suggest the integration of break-out chat rooms. In our findings, socialising with others was a comparatively underrepresented motivation; it may be that passive consumers are overwhelmed by large chat rooms populated by various communities. Further, large tournament livestreams—which may accrue hundreds of thousands of viewers—often produce a chaotic, fast-paced live chat conducive to meaningful interaction or conversation. Implementing smaller break-out chat rooms within streams that allow spectators to find others with common interests may promote more engagement with interactive features.

Finally, we also note that participants also reported viewing streams for “Learning” purposes: translating learnings—acquired from watching high-level, professional play—to their own gameplay. This motivation may represent one avenue for encouraging spectator engagement. Current spectator features exist to facilitate this motivation such as Pro View (i.e., a feature that provides each player’s point of view along with their stats); however, such features are often limited to those who pay a subscription fee. Providing an open access feature to individual player’s Point-Of-View (POVs) in livestreaming platforms may promote engagement such as increase their usage of interactive features among those motivated to improve their gameplay.

## 6 CONCLUSION

In summary, we report an exploratory investigation into the motivations behind why spectators watch live streams of professional esports tournaments.

Participants identified a variety of motivations for viewing esports tournament livestreams, as well as motivations for engaging with interactive livestream features. Broadly, spectators were motivated to watch tournament livestreams by the competitive environment, the presence of one or more preferred teams or players, the potential for translating knowledge to their own gameplay, and the entertainment value associated with the unpredictability or exciting nature of a professional match. Among spectators who also engage with interactive stream features (such as live chat), we find that the most prominent motivations are the social aspects, entertainment, and incentives for participation. Further, our findings imply that offering extrinsic rewards—such as in-game items—may also encourage spectator engagement.

Future research might consider leveraging these motivations in service of creating a spectator typology, which will help designers and stakeholders better understand their target audience. This knowledge represents an initial step towards equipping developers with the tools and knowledge to tailor engaging and positive spectator experiences.

## REFERENCES

- [1] Richard Bartle. 2014. HEARTS , CLUBS , DIAMONDS , SPADES .: June 1996 (2014).
- [2] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (1 2006), 77–101. <https://doi.org/10.1191/1478088706qp063oa>
- [3] Luis Javier Cabeza-Ramírez, Sandra M. Sánchez-Cañizares, and Fernando J. Fuentes-García. 2020. Motivations for the use of video game streaming platforms: The moderating effect of sex, age and self-perception of level as a player. *International Journal of Environmental Research and Public Health* 17, 19 (2020), 1–20. <https://doi.org/10.3390/ijerph17197019>

- [4] Claudia Flores-Saviaga, Joseph Seering, Jessica Hammer, Stuart Reeves, Juan Pablo Flores, and Saiph Savage. 2019. Audience and streamer participation at scale on twitch. *HT 2019 - Proceedings of the 30th ACM Conference on Hypertext and Social Media 2* (2019), 277–278. <https://doi.org/10.1145/3342220.3344926>
- [5] Colin Ford, Dan Gardner, Leah Elaine Horgan, Calvin Liu, a. m. Tsaasan, Bonnie Nardi, and Jordan Rickman. 2017. Chat Speed OP PogChamp: Practices of Coherence in Massive Twitch Chat. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '17)*. Association for Computing Machinery, New York, NY, USA, 858–871. <https://doi.org/10.1145/3027063.3052765>
- [6] Guo Freeman and Donghee Yvette Wohn. 2017. eSports As An Emerging Research Context at CHI: Diverse Perspectives on Definitions. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '17)*. ACM, New York, NY, USA, 1601–1608. <https://doi.org/10.1145/3027063.3053158>
- [7] Guo Freeman and Donghee Yvette Wohn. 2017. Social support in eSports: Building emotional and esteem support from instrumental support interactions in a highly competitive environment. *CHI PLAY 2017 - Proceedings of the Annual Symposium on Computer-Human Interaction in Play* (2017), 435–447. <https://doi.org/10.1145/3116595.3116635>
- [8] Juho Hamari and Max Sjöblom. 2017. What is eSports and why do people watch it? *Internet Research* 27, 2 (2017), 211–232. <https://doi.org/10.1108/IntR-04-2016-0085>
- [9] Zorah Hilvert-Bruce, James T Neill, Max Sjöblom, and Juho Hamari. 2018. Social motivations of live-streaming viewer engagement on Twitch. *Computers in Human Behavior* 84 (2018), 58–67. <https://doi.org/10.1016/j.chb.2018.02.013>
- [10] Andrzej Marczewski. 2016. Marczewski's Player and User Types Hexad. *Even Ninja Monkeys Like to Play: Gamification, Game Thinking and Motivational Design* 24, 3 (2016), 65–80.
- [11] Ilya Musabirov, Denis Bulygin, Paul Okopny, and Ksenia Konstantinova. 2018. Event-driven spectators' communication in massive eSports online chats. *Conference on Human Factors in Computing Systems - Proceedings 2018-April* (2018), 1–6. <https://doi.org/10.1145/3170427.3188447>
- [12] Florian Neus, Frederic Nimmermann, Katja Wagner, and Hanna Schramm-Klein. 2019. Differences and similarities in motivation for offline and online eSports event consumption. *Proceedings of the Annual Hawaii International Conference on System Sciences 2019-Janua* (2019), 2458–2467. <https://doi.org/10.24251/hicss.2019.296>
- [13] Newzoo. 2021. *Newzoo's Global Esports & Live Streaming Market Report 2021*. Technical Report. [https://newzoo.com/insights/trend-reports/newzoos-global-esports-live-streaming-market-report-2021-free-version/?utm\\_campaign=GEMR2021&utm\\_source=oldercontentto2021freereport&utm\\_content=freereport](https://newzoo.com/insights/trend-reports/newzoos-global-esports-live-streaming-market-report-2021-free-version/?utm_campaign=GEMR2021&utm_source=oldercontentto2021freereport&utm_content=freereport)
- [14] Raquel Robinson, Zachary Rubin, E M Elena Márquez Segura, and Katherine Isbister. 2017. All the Feels: Designing a Tool That Reveals Streamers' Biometrics to Spectators. In *Proceedings of the 12th International Conference on the Foundations of Digital Games (FDG '17, Vol. Part F1301)*. ACM, New York, NY, USA, 36:1–36:6. <https://doi.org/10.1145/3102071.3102103>
- [15] Damian Schofield, Robert Ledone, Suny Oswego, and New York. 2020. The Motivations of a Video Game Streamers and their Viewers. December 2019 (2020).
- [16] Joseph Seering, Saiph Savage, Michael Eagle, Joshua Churchin, Rachel Moeller, Jeffrey P Bigham, and Jessica Hammer. 2017. Audience Participation Games: Blurring the Line Between Player and Spectator. In *Proceedings of the 2017 Conference on Designing Interactive Systems (DIS '17)*. ACM, New York, NY, USA, 429–440. <https://doi.org/10.1145/3064663.3064732>
- [17] Samantha Stahlke, James Robb, and Pejman Mirza-Babaei. 2018. The fall of the fourth wall: Designing and evaluating interactive spectator experiences. *International Journal of Gaming and Computer-Mediated Simulations* 10, 1 (2018), 42–62. <https://doi.org/10.4018/IJGCM.2018010103>
- [18] Alina Striner, Sasha Azad, and Chris Martens. 2019. A spectrum of audience interactivity for entertainment domains. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)* 11869 LNCS, April (2019), 214–232. [https://doi.org/10.1007/978-3-030-33894-7\\_23](https://doi.org/10.1007/978-3-030-33894-7_23)
- [19] Nick Yee. 2006. Motivations for play in online games. *Cyberpsychology & behavior : the impact of the Internet, multimedia and virtual reality on behavior and society* 9, 6 (12 2006), 772–775. <https://doi.org/10.1089/cpb.2006.9.772>
- [20] Nick Yee and Jeremy Bailenson. 2007. The proteus effect: The effect of transformed self-representation on behavior. *Human Communication Research* 33, 3 (2007), 271–290. <https://doi.org/10.1111/j.1468-2958.2007.00299.x>
- [21] Wen Bo Yu, James Robb, and Pejman Mirza-Babaei. 2020. Profiling livestream spectators. *CHI PLAY 2020 - Extended Abstracts of the 2020 Annual Symposium on Computer-Human Interaction in Play 2* (2020), 403–407. <https://doi.org/10.1145/3383668.3419904>
- [22] Franziska Zimmer and Katrin Scheibe. 2019. What drives streamers? Users' characteristics and motivations on social live streaming services. *Proceedings of the Annual Hawaii International Conference on System Sciences 2019-Janua* (2019), 2538–2547. <https://doi.org/10.24251/hicss.2019.306>